

The easiest dice to tell apart

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Abstract

Two loaded dice have k faces, and no face of either die comes up with probability more than r . Both are rolled n times, and we must guess which die produced the sequence. We find the pair that is easiest to tell apart, for every k , r and n : a staircase $(r, \dots, r, 1 - mr, 0, \dots, 0)$, $m = \lfloor 1/r \rfloor$, against its own reversal. This answers a question on Mathematics Stack Exchange, which asked for $k = 4$ and $1/3 < r < 1/2$. For that case we also find every pair that does as well, except when $n = 2$, where we exhibit a one-parameter family. The proof compares the dice face set by face set at every exchange rate, which takes three lines for one roll; a standard argument then handles n rolls. Every step is checked in the Lean proof assistant.

1 The question

Let $p = (p_1, \dots, p_k)$ and $q = (q_1, \dots, q_k)$ be the face probabilities of two dice. Roll one of them n times. A sequence of faces $w = (i_1, \dots, i_n)$ has probability $P(w) = p_{i_1} \cdots p_{i_n}$ under the first die and $Q(w) = q_{i_1} \cdots q_{i_n}$ under the second. The *overlap*

$$\text{ov}_n(p, q) = \sum_w \min(P(w), Q(w))$$

measures how hard the dice are to tell apart. If each die is chosen with probability $\frac{1}{2}$, the best possible guess is wrong with probability exactly $\frac{1}{2} \text{ov}_n(p, q)$: for each w , guess the die that makes w more likely, and you are wrong with probability $\frac{1}{2} \min(P(w), Q(w))$.

Identical dice have overlap 1, and dice with no face in common have overlap 0. The question [1] puts a cap on how loaded the dice may be: every p_i and every q_i is at most r . For $k = 4$ and $\frac{1}{3} < r < \frac{1}{2}$, how small can the overlap be? The asker proved $\text{ov}_n \geq (2 - 4r)^n$, and computed the overlap of the pair

$$p = (r, s, r, 0), \quad q = (0, r, s, r), \quad s = 1 - 2r,$$

which bounds the least overlap from above. At $n = 2$ and $r = \frac{2}{5}$ the two bounds are $\frac{4}{25}$ and $\frac{6}{25}$. Stop and try to decide which is the truth before reading on.

The pair is the answer, and it is a case of a pattern that holds for every number of faces and every cap. Write $m = \lfloor 1/r \rfloor$ and define the *staircase* and its reversal

$$p^* = (\underbrace{r, \dots, r}_m, 1 - mr, 0, \dots, 0), \quad q_i^* = p_{k+1-i}^*.$$

(When $1/r$ is an integer, $1 - mr = 0$.) For $k = 4$ and $\frac{1}{3} < r < \frac{1}{2}$ this is $p^* = (r, r, s, 0)$ and $q^* = (0, s, r, r)$, the asker's pair with two faces renamed (Figure 1).

Theorem 1. *Let $kr \geq 1$, so that capped dice exist. For every $n \geq 1$ and every choice of capped dice, which may even differ from roll to roll, the overlap of the n rolls is at least $\text{ov}_n(p^*, q^*)$.*

Corollary 1. For $k = 4$ and $\frac{1}{3} < r < \frac{1}{2}$ the least overlap is the asker's bound,

$$C_n(r) = 2 \sum_{j > n/2} \binom{n}{j} s^j r^{n-j} + [n \text{ even}] \binom{n}{n/2} (rs)^{n/2}.$$

In particular $C_2(\frac{2}{5}) = \frac{6}{25}$.

The formula is the asker's evaluation of their pair [1], which we have checked for $n \leq 6$; it comes from sorting the words of the staircase pair: a word containing face $(r, 0)$ or face $(0, r)$ contributes nothing, and a word made of j faces (s, r) and $n - j$ faces (r, s) contributes $\min(s^j r^{n-j}, r^j s^{n-j})$.

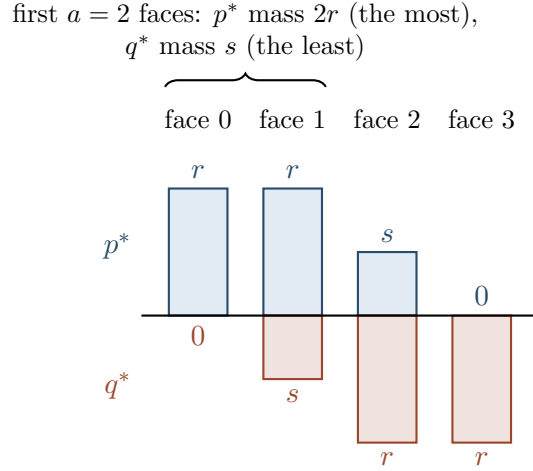


Figure 1: The staircase pair for $k = 4$, $r = \frac{2}{5}$, $s = \frac{1}{5}$: p^* above the line, q^* below. Any two faces carry at most $2r$ of a capped die. Any two faces carry at least $1 - 2r = s$ of a capped die, because the other two carry at most $2r$. The first two faces of the staircase pair meet both limits at once. The same holds for the first a faces, for every a .

2 Overlap as an excess

For $a, b \geq 0$, $\min(a, b) = a - (a - b)^+$, where $x^+ = \max(x, 0)$. Since the $P(w)$ add up to 1,

$$\text{ov}_n = 1 - \sum_w (P(w) - Q(w))^+.$$

So a small overlap means a large *excess* of P over Q . The proof needs the excess at every exchange rate, so for $u, v \geq 0$ set

$$g_{p,q}(u, v) = \sum_{i=1}^k (u p_i - v q_i)^+.$$

Read it as a bet on faces. Choose a set A of faces; gain u for each unit of p on A and lose v for each unit of q on A . The best set is $A = \{i : u p_i > v q_i\}$, so

$$g_{p,q}(u, v) = \max_A (u p(A) - v q(A)), \quad p(A) = \sum_{i \in A} p_i. \quad (1)$$

For n rolls, with die $(p^{(j)}, q^{(j)})$ at roll j , define $G(u, v) = \sum_w (u P(w) - v Q(w))^+$ in the same way. The overlap is $1 - G(1, 1)$.

3 One roll

Lemma 1. *Let p and q be capped dice and let A be a set of a faces. Then*

$$p(A) \leq \min(1, ar), \quad q(A) \geq \max(0, 1 - (k - a)r).$$

The first a faces of the staircase pair meet both bounds: $p^(\{1, \dots, a\}) = \min(1, ar)$ and $q^*(\{1, \dots, a\}) = \max(0, 1 - (k - a)r)$.*

Proof. Each face carries at most r , and the whole die carries 1; this gives the bound on $p(A)$. The other $k - a$ faces carry at most $(k - a)r$ of q , which gives the bound on $q(A)$. For the staircase, the partial sums of p^* are $r, 2r, \dots, mr$, then 1 for ever, which is $\min(1, ar)$. The first a faces of q^* are the last a faces of p^* , which carry $1 - \min(1, (k - a)r) = \max(0, 1 - (k - a)r)$. $\square$

Proposition 1. *For capped dice p, q and all $u, v \geq 0$, $g_{p,q}(u, v) \leq g_{p^*,q^*}(u, v)$.*

Proof. Take the best set A in (1) and let $a = |A|$. By Lemma 1, the bet on A gains at most $u \min(1, ar) - v \max(0, 1 - (k - a)r)$. The staircase pair gains exactly that by betting on its first a faces, and its best bet gains at least as much. $\square$

The staircase pair wins at every exchange rate at once, and that is what the next step needs: when we go from one roll to many, each roll is judged at a rate that the other rolls set.

4 Many rolls

Split a word into its first face i and the rest x . Then $P(w) = p_i P'(x)$ and $Q(w) = q_i Q'(x)$, where P', Q' belong to the last $n - 1$ rolls, and the excess can be summed in either order:

$$G(u, v) = \sum_i G'(u p_i, v q_i) = \sum_x g_{p,q}(u P'(x), v Q'(x)). \quad (2)$$

The first form says the last $n - 1$ rolls are judged at the rate $(u p_i, v q_i)$ that the first roll sets; the second says the first roll is judged at the rate $(u P'(x), v Q'(x))$ that the others set.

Proof of Theorem 1. We show $G(u, v) \leq G^*(u, v)$ for all $u, v \geq 0$, where G^* belongs to n rolls of the staircase pair, by induction on n . For $n = 0$ both sides are $(u - v)^+$. For the step, use the first form of (2) and the induction hypothesis to replace the last $n - 1$ rolls by staircase rolls; then use the second form and Proposition 1, at each rate $(u P'(x), v Q'(x))$, to replace the first roll (Figure 2). Neither step decreases the excess. At $u = v = 1$ this is the theorem. $\square$

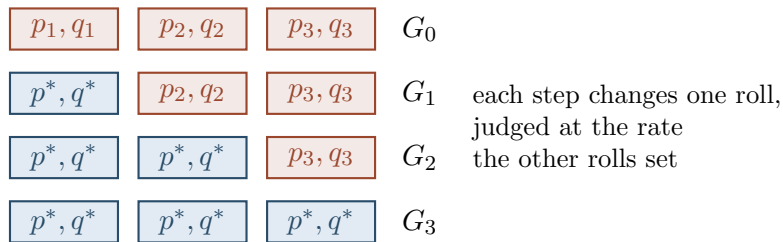


Figure 2: Three rolls, with excesses $G_0 \leq G_1 \leq G_2 \leq G_3$. Replacing one roll by the staircase pair cannot decrease the excess, because the staircase pair wins at every rate (Proposition 1).

Remark. This second step is standard. Proposition 1 says that (p^*, q^*) is more informative than every capped pair in Blackwell's sense: domination at every rate is equivalent to one experiment being a garbling of the other (Blackwell [2, Theorem 10], in the form of [3, Theorem 2.5]), and the induction above is the familiar fact that products keep that order. The same pattern of one dominant experiment, then its k -fold product, is how Kairouz, Oh and Viswanath prove their optimal composition theorem for differential privacy [3, Section 3.1]. What is particular here is that the capped dice have a dominant pair at all, and Lemma 1 shows why: one chain of nested sets is extreme for both dice at once.

5 Who else does as well, for four faces

From here on $k = 4$, $\frac{1}{3} < r < \frac{1}{2}$ and $s = 1 - 2r$, so $0 < s < r$. Relabelling faces does not change the overlap, so every pair with one face of each kind $(r, 0)$, (r, s) , (s, r) , $(0, r)$ reaches $C_n(r)$.

Theorem 2. *Let p, q be capped dice with four faces and $\text{ov}_n(p, q) = C_n(r)$.*

- (a) *For $n = 1$ this happens exactly when $\max(p_i, q_i) = r$ for every face.*
- (b) *For $n \geq 3$ it happens only for the four kinds of face above, one of each.*

For $n = 2$ it also happens for $p = (r, r, x, s - x)$, $q = (0, s, r, r)$ with $0 \leq x \leq s$.

Part (a) is a line of algebra: $\text{ov}_1 = \sum_i \min(p_i, q_i) = 2 - \sum_i \max(p_i, q_i) \geq 2 - 4r$. The family for $n = 2$ is a direct computation of $G(1, 1)$ through (2); it equals $r + 2r^2 - s^2 + rs$ for every x . We have not proved that this family, with its relabellings and with p and q swapped, is the whole answer for $n = 2$.

For (b), equality forces equality all the way along the chain in the proof of Theorem 1. Since each step there compares sums term by term, every term must agree.

Lemma 2. *If $\text{ov}_n(p, q) = C_n(r)$, then $g_{p,q} = g_{p^*,q^*}$ at the rate $(P^*(x), Q^*(x))$ for every word x of $n - 1$ staircase rolls, and at $(p_i P^*(x), q_i Q^*(x))$ for every face i and every word x of $n - 2$ staircase rolls.*

The next lemma says what equality at one rate means. Write $B(a) = u \min(1, ar) - v \max(0, 1 - (4 - a)r)$ for the bound on a bet over a faces from Lemma 1. With four faces the five values are

$$B(0) = 0, \quad B(1) = ur, \quad B(2) = 2ur - vs, \quad B(3) = u - v(1 - r), \quad B(4) = u - v.$$

Lemma 3. *If $u, v > 0$ and $g_{p,q}(u, v) = g_{p^*,q^*}(u, v)$, let $A = \{i : u p_i > v q_i\}$ and $a = |A|$. Then $B(a)$ is the largest of the five values, $p(A) = \min(1, ar)$ and $q(A) = \max(0, 1 - (4 - a)r)$.*

Proof. In the proof of Proposition 1 every inequality must be an equality, and the two inequalities of Lemma 1 are each multiplied by a positive number. $\square$

Proof of Theorem 2(b). Write $t = v/u$. Three rates pin the pair down.

A high rate. The word of $n - 1$ faces (s, r) gives $t = (r/s)^{n-1} > r/s$. Then $B(1)$ is the unique largest value, so $|A| = 1$, $p(A) = r$ and $q(A) = 0$: some face is $(r, 0)$.

A low rate. The word of $n - 1$ faces (r, s) gives $t = (s/r)^{n-1} < s/r$. Now $B(3)$ is the unique largest, so $p(A) = 1$ and $q(A) = 1 - r$. The face outside A is $(0, r)$.

A middle rate. The other two faces carry $1 - r$ of each die, so each of their p and q values lies between s and r , and one of them, say (p_i, q_i) , has $p_i \geq q_i$.

If n is odd, the word with $\frac{n-1}{2}$ faces of each kind (r, s) , (s, r) gives $t = 1$. Then $B(2)$ is the unique largest, so A is two faces with $p = r$ on both and $q(A) = s$. Face $(r, 0)$ is one of them, so the other is (r, s) , and the remaining face is (s, r) .

If n is even, use the face i with $p_i \geq q_i$ and the balanced word of $n - 2$ rolls: this gives $t = q_i/p_i$, which lies between s/r and 1. There $B(2)$ is largest, tied with $B(3)$ only when $t = s/r$. If $|A| = 3$, then $q_i/p_i = s/r$ with $p_i \leq r$ and $q_i \geq s$, so $(p_i, q_i) = (r, s)$. If $|A| = 2$, then A has $p = r$ on both faces and $q = r$ on both faces outside A ; it contains the face $(r, 0)$ and not the face $(0, r)$. Suppose face i were outside A . Then $q_i = r$, hence $p_i = r$, and the fourth face would be the second face of A , with $p = r$; but the four values of p would then add up to $3r > 1$. So $A = \{(r, 0), i\}$ and $q_i = s - 0 = s$. Either way face i is (r, s) , and the fourth face is $(1 - 2r, 1 - r - s) = (s, r)$. $\square$

Verification. Every theorem and lemma is proved in Lean 4 with Mathlib [4], using only the standard axioms: Lemma 1 and Proposition 1 for every k and r , the product step for dice that change from roll to roll, Corollary 1 as the statement that the asker’s pair has the least overlap, Lemmas 2 and 3, and Theorem 2 with its $n = 2$ family. The closed form in Corollary 1 is checked by program, not in Lean. Python programs check, in exact rational arithmetic, the one-roll domination for $k \leq 7$ at every breakpoint, the many-roll inequality for random mixed dice, the closed form in Corollary 1 for $n \leq 6$, and the equality cases for $n \leq 4$; deliberately wrong variants fail each check. An independent program, sharing no code with the others, searches numerically for pairs with a smaller overlap and finds none. Everything is at <https://github.com/jvvk/mathematics/tree/main/easiest-dice>.

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